ai-protect ↔ ai-gateway — information flow

Supersedes the flow-diagram content of the older design record linked from `ai-gateway/docs/README.md` (`claude.ai/code/artifact/9d348c21-...`) — that page's own header already flags itself as predating the module split (three now-retired CLI modes, an old bootstrap flow). This doc reflects the current tree; re-check it against the code before relying on a detail, the same caveat the old one carried.

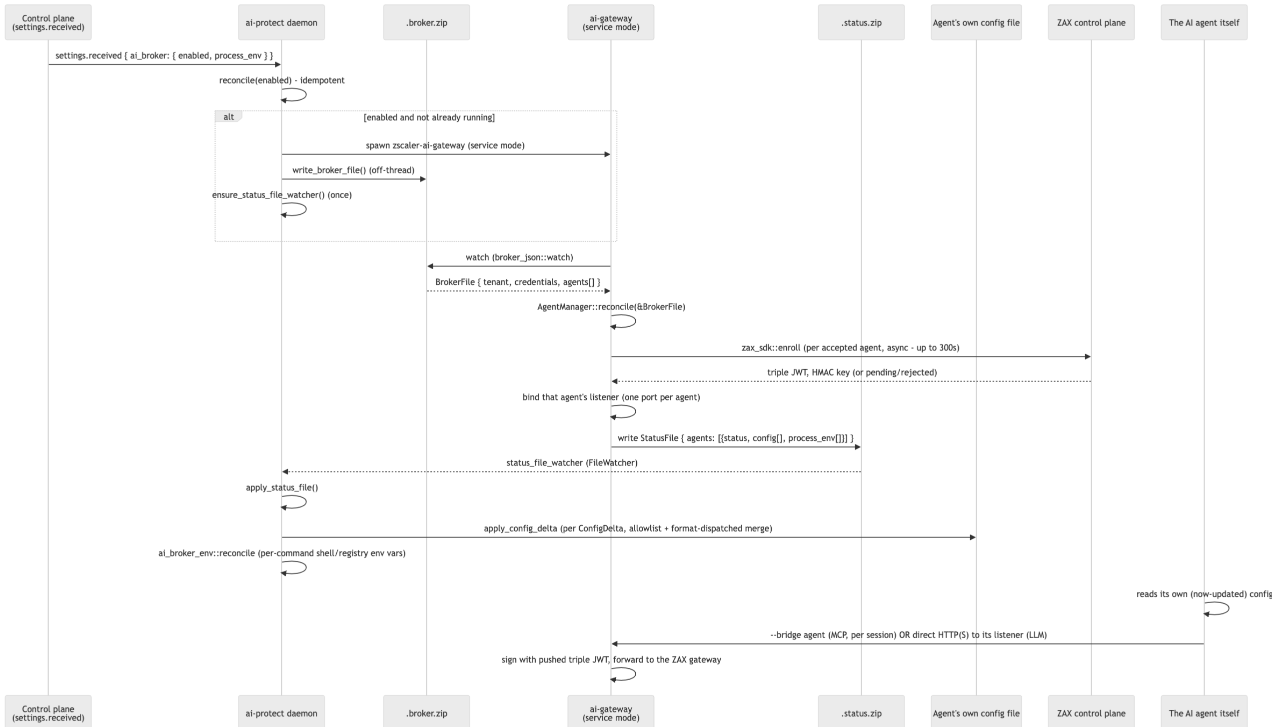
Companion docs: `broker-status-file-schema.md` (the wire schema referenced throughout this doc) and `ai-gateway/docs/OperationalModes.md` (the operating guide — how to run and verify each mode).

The two processes

	ai-protect	ai-gateway (`zscaler-ai-gateway`)
Privilege	root / LocalSystem, resident daemon	user-space, per-session or one long-lived "service mode" instance
Build graph	never links ai-gateway's crates	separate binary, shipped alongside the daemon
Owns	discovery, policy, the real agent config files	enrolment, MCP/LLM listeners, the credential store
Talks to	ai-gateway (files only), ai-platform (its own device-plane channel)	the ZAX control plane (vector), the ZAX gateway (Optimus)

No socket, no named pipe, between the two. Each writes exactly one file for the other to watch; either can restart independently, and a stale watcher (wrong binary version on one side) fails silently rather than erroring — see the Gotchas section.

Lifecycle, end to end



Four things independently wake ai-gateway's service-mode loop (not just `.broker.zip` changing) — this is easy to under-model if you only look at the happy path:

Wake source	Why it exists
`.broker.zip` changed	the agent list or the pushed credential moved
an enrolment task completed	`zax_sdk::enroll` blocks up to 300s on approval — can't be awaited inline without stalling every other write
the credential store changed	something *other than* this process wrote it (tamper) — restore
a 30s refresh tick	the credential's freshness and an agent's triple-JWT validity both age out with nothing else writing a file

On the ai-protect side, the corresponding independent triggers are: `settings.received` (the `ai_broker` toggle), an agent-discovery rescan (`on_agent_rescan` — rewrites `.broker.zip`'s agent list, no diffing), and the `.status.zip` file watcher itself.

Two legs, two mechanisms

MCP leg — `--bridge <agent>` is a separate CLI mode, spawned by the *agent itself* per session (not by the daemon, not part of service mode). It reads the credential store ai-gateway's service mode already populated, relays JSON-RPC over stdio, and signs outbound gateway calls with the pushed triple JWT. With no record for that agent yet, the bridge still starts and advertises no tools — every request then fails `NoCredentials` rather than an unsigned request reaching the gateway.

LLM leg — one in-process listener per agent (`http` or `https`, picked by that adapter's config), all behind one request pipeline (`ai-gateway-listener`): resolve destination → build the outbound request in one place (strip hop-by-hop + any caller-supplied `signature*` / `x-zax-*` headers, set `Host`, sign, insert the four ZAX headers) → dial the configured gateway over an owned socket (HTTP/1.1 only — the `Host` header carries the real destination, which HTTP/2's `:authority` would contradict). `https` serves **both** a base-URL client and a CONNECT-then-TLS forward-proxy client **on the same port**, disambiguated by peeking the first byte (`0x16` = TLS ClientHello, ASCII `C` = a CONNECT request line).

Config delivery — the one-writer rule

`ConfigDelta` / `ManagedEnvVar` are *descriptions* of intent; ai-gateway never writes an agent's real config file or touches the OS environment itself. ai-protect is the sole writer, and it enforces two independent gates before any byte lands (detailed in the schema doc): the file must be on `ALLOWED_CONFIG_PATHS`, and the merge is dispatched by extension (TOML/YAML/ JSON), each format refusing rather than replacing on a parse failure.

Full snapshot in, idempotent merge out. `.status.zip` re-sends every serviceable agent's complete wiring on every write (not a diff), so ai-protect's merge has to tolerate being handed the same delta repeatedly — a JSON-object merge and an atomic file rewrite are naturally idempotent, so this costs nothing, but it's worth knowing when reading logs: "applied config delta ... for agent X" is expected on nearly every `.status.zip` write for a healthy agent, not just the first time.

No config until the agent can be served. Empty `config: []` is "nothing to change," not "unwire" — there is no retract verb in this protocol. An agent whose triple JWT lapses keeps whatever config it was last given; its requests fail `NoCredentials` until the credential refreshes. Undoing a wiring is `--recover`'s job (CLI-only today, not wired into the disable path — see Gaps below).

Known gaps worth flagging on review

Carried over verbatim from the current code's own doc comments — not guesses, and not fixed by this doc:

1. **Privilege drop is incomplete on Linux.** Windows and macOS both drop to the console user's session before spawning ai-gateway; Linux has no portable console-session-uid primitive without parsing `utmp` or shelling to `loginctl` (unacceptable in a root daemon), so a Linux root daemon still spawns ai-gateway in its own root context.
2. **The user-JWT identity handoff is a stub.** `identity::zax_user_credential::fetch` on the ai-protect side returns nothing real yet; `.broker.zip`'s `credentials` block is blank in practice, so ai-gateway always falls back to its own cached refresh token or interactive OIDC.
3. **Single-user scoping.** `.broker.zip` / `.status.zip` watch/write is scoped to one supervised broker's home. Multi-user hosts are not yet handled by this file-protocol migration.

4. **Disabling the broker doesn't unwire config.** `reconcile(false)` stops the process and tears down process-env vars (`ai_broker_env::reconcile(&empty)`), but the `ConfigDelta`/`--recover` path is CLI-only — `unwire_other_agents()`/`recover()` are never called from the disable path. A device that turns `ai_broker.enabled` off keeps whatever config deltas were last applied until someone runs `--recover` by hand.
5. **No snapshot-before-merge on the ai-protect side.** The old bootstrap captured an agent's prior `ANTHROPIC_BASE_URL` before overwriting it, so `--recover` could restore it. `apply_config_delta` merges atomically but takes no backup and records no prior value — recorded as a tracked follow-up in `ai-gateway/docs/OperationalModes.md`, not fixed here.
6. **The ZAX root CA is embedded in the binary and trusted unconditionally**, release builds included — whoever holds its private key can present a certificate the gateway leg accepts, on every device running this binary. Narrowing this needs gating `trust::ZAX_ROOT_PEM` on a tenant/environment signal.
7. **Loopback is the only gate on a listener port.** Any local account on the box can reach it and have traffic signed with this device's ZAX identity. The per-install path-token that used to narrow this was deliberately dropped (see `ai-gateway/listener/DESIGN.md`).

Where the two `advisory`/older concepts went

Two things a reviewer who saw the earlier design record may look for and not find, on purpose:

- **`enforce_mcp` advisories** (`<agent config dir>/advisory.json`, mirroring ai-protect's resolved MCP allow/block decision into the agent's directory for visibility) — removed. The current `StatusFile`/`BrokerFile` schema has no `advisories` field, so nothing on either side carries or writes it today. Restorable, but both halves (the writer and the schema field) need to come back together, not just one.
- **Per-install path token in the listener URL** (`https://127.0.0.1:<port>/<token>/v1`) — removed; see gap 7 above.